

# **BUSINESS ANALYTICS INDIVIDUAL CASE STUDY**

## **Beyond the Headline Rate: A Data-Driven Transparency Scoring Framework for Advertised vs. Effective Interest Rates in the Indian Personal Loan Market**

**Student Name:** \_\_\_\_\_

**Register Number:** \_\_\_\_\_

**Class / Section:** \_\_\_\_\_

**Data collection date:** 25 September 2026

**Primary source:** Paisabazaar personal-loan comparison page

**Submission note:** The supplied proposal contains a different student's identity. Replace these placeholders before submission.

## **1. Problem Statement and Objectives**

Advertised personal-loan rates do not by themselves describe the full borrowing cost. Processing fees and applicable GST can be deducted from the sanctioned amount while EMI is calculated on the sanctioned principal. This creates a difference between the headline rate and the effective annualized cost of financing.

This case study standardizes published lender terms into comparable loan scenarios and calculates an effective APR from the net amount disbursed and EMI stream. It is designed as an analytical screening framework, not as a legal or regulatory determination about lender conduct.

### **Objectives**

- Quantify the gap between published advertised rate and scenario-level effective APR.
- Examine how processing-fee burden, loan amount and tenure are associated with the gap.
- Compare public banks, private banks and NBFCs.
- Create a lender-level transparency score based on gap magnitude and consistency.

## 2. Data Collection and Dataset Description

The primary source is the publicly accessible Paisabazaar personal-loan comparison page. It provides lender-level interest-rate, loan-amount, tenure and processing-fee information. The source states that rates are indicative and that final offers depend on borrower profile.

The dataset contains 13,024 standardized analytical scenarios derived from 32 lender records. Each lender is crossed with 37 loan amounts from ₹1,00,000 to ₹10,00,000 and 11 tenures from 12 to 84 months. Therefore  $32 \times 37 \times 11 = 13,024$ .

**Critical distinction:** these are scenario observations, not independent scraped advertisements. The raw source-derived lender table is included separately.

### Key variables

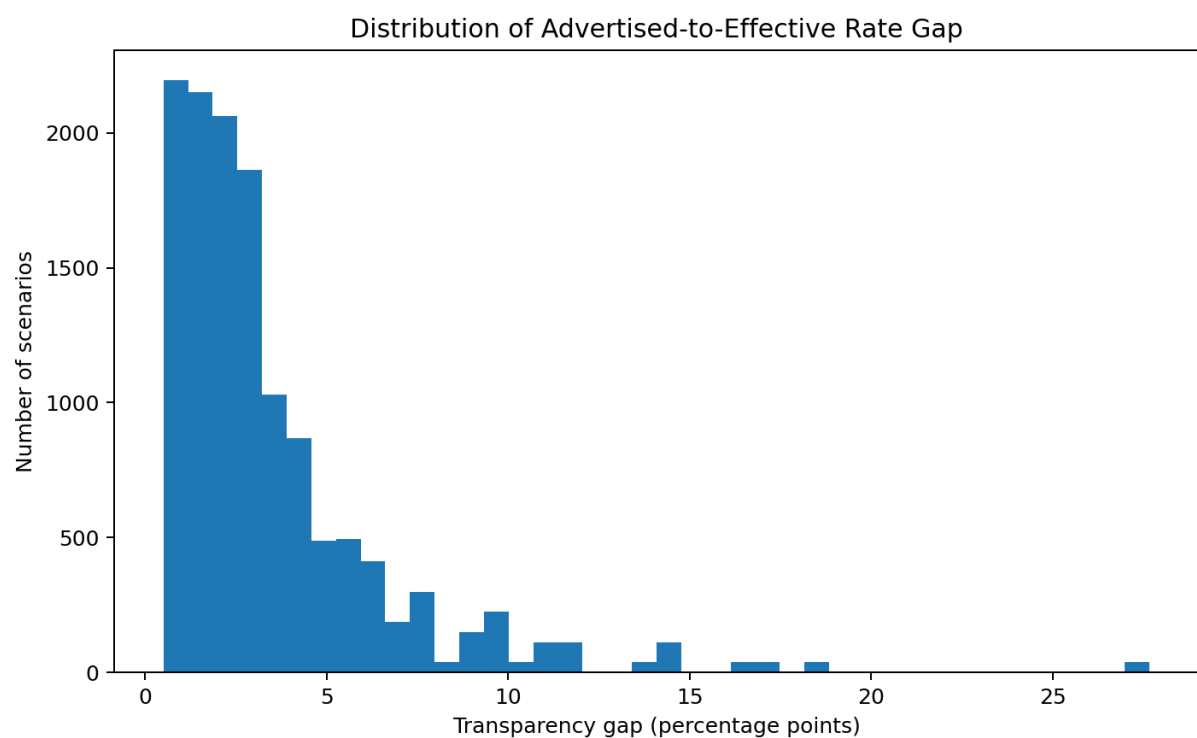
| Attribute                 | Description                                    |
|---------------------------|------------------------------------------------|
| lender_name               | Bank/NBFC name                                 |
| lender_type               | Public Bank, Private Bank or NBFC              |
| loan_amount_inr           | Standardized sanctioned amount                 |
| tenure_months             | Standardized repayment period                  |
| advertised_rate_pct       | Published starting/lower-bound rate used       |
| processing_fee_inr        | Fee calculated from published rule             |
| gst_on_processing_fee_inr | 18% GST on processing fee                      |
| net_disbursed_inr         | Sanctioned amount less fee and GST             |
| emi_inr                   | EMI calculated from advertised nominal rate    |
| effective_apr_pct         | APR implied by net disbursement and EMI stream |
| transparency_gap_pp       | Effective APR minus advertised rate            |

### 3. Data Preparation and Exploratory Analysis

The dataset was checked for missing values, duplicates, lender counts and variable types. Processing-fee burden was derived as processing fee divided by sanctioned amount. Loan amount and tenure bands were created for interpretation. The analysis uses the source-derived lender terms as fixed inputs and does not claim borrower-level representativeness.

#### Descriptive statistics

| Metric              | Mean       | Median     | Std.       | Min        | Max          |
|---------------------|------------|------------|------------|------------|--------------|
| Advertised rate (%) | 10.70      | 10.00      | 1.82       | 8.00       | 18.00        |
| Effective APR (%)   | 14.19      | 13.24      | 3.90       | 8.50       | 39.67        |
| Gap (pp)            | 3.50       | 2.58       | 3.18       | 0.50       | 27.67        |
| Processing fee (■)  | 15,059.57  | 9,500.00   | 14,065.85  | 500.00     | 90,000.00    |
| EMI (■)             | 20,054.53  | 15,848.61  | 15,882.79  | 1,558.62   | 91,679.99    |
| Total repayment (■) | 668,176.90 | 662,862.44 | 332,330.14 | 104,386.11 | 1,765,498.39 |



The mean transparency gap is 3.50 percentage points and the median is 2.58. The right tail reflects scenarios where one-time fees have a relatively large annualized effect.

## 4. Analytics Method and Implementation

For each scenario, the processing fee is calculated from the published fee rule and 18% GST is applied to that fee. Net disbursement is the sanctioned amount less fee and GST.

**Net disbursement = Loan amount – Processing fee – GST on processing fee**

EMI is calculated from the published nominal annual rate using the reducing-balance EMI formula. The effective monthly IRR is then solved from the net disbursement and EMI cash flows.  $\text{Effective APR} = (1 + \text{monthly IRR})^{12} - 1$ . The transparency gap is  $\text{Effective APR} - \text{Advertised rate}$ .

The IRR approach is preferable to simply adding the fee percentage to the advertised rate because the fee changes the amount received at time zero.

### Statistical method

The lender-type comparison uses a Kruskal-Wallis test on one median-gap observation per lender. This avoids treating the deterministic scenario expansion as thousands of independent lenders. An exploratory OLS model uses scenario observations with standard errors clustered by lender.

5. Comparison with State-of-the-Art Methods

| Study / year                 | Dataset                                 | Method                                   | Key result / evaluation                                             | Comparison                                                    |
|------------------------------|-----------------------------------------|------------------------------------------|---------------------------------------------------------------------|---------------------------------------------------------------|
| Tantri & Vishen (2025)       | Indian secured corporate loan contracts | Quasi-experimental transparency analysis | Studies disclosure of bank cost benchmarking                        | Qualitative lender-level setting                              |
| Ali & Marisetty (2026 issue) | ~2.19M reviews from 110 Indian EBLs     | EDM + Poisson regression                 | Analyzes borrower experience and negative lender experiences        | Borrower-level experience focuses on pricing/fee transparency |
| Sidharta et al. (2024)       | 272 fintech borrowers                   | Questionnaire + quantitative model       | Tests interest rate, administrative fee and price effects over time | Qualitative lender-level setting                              |

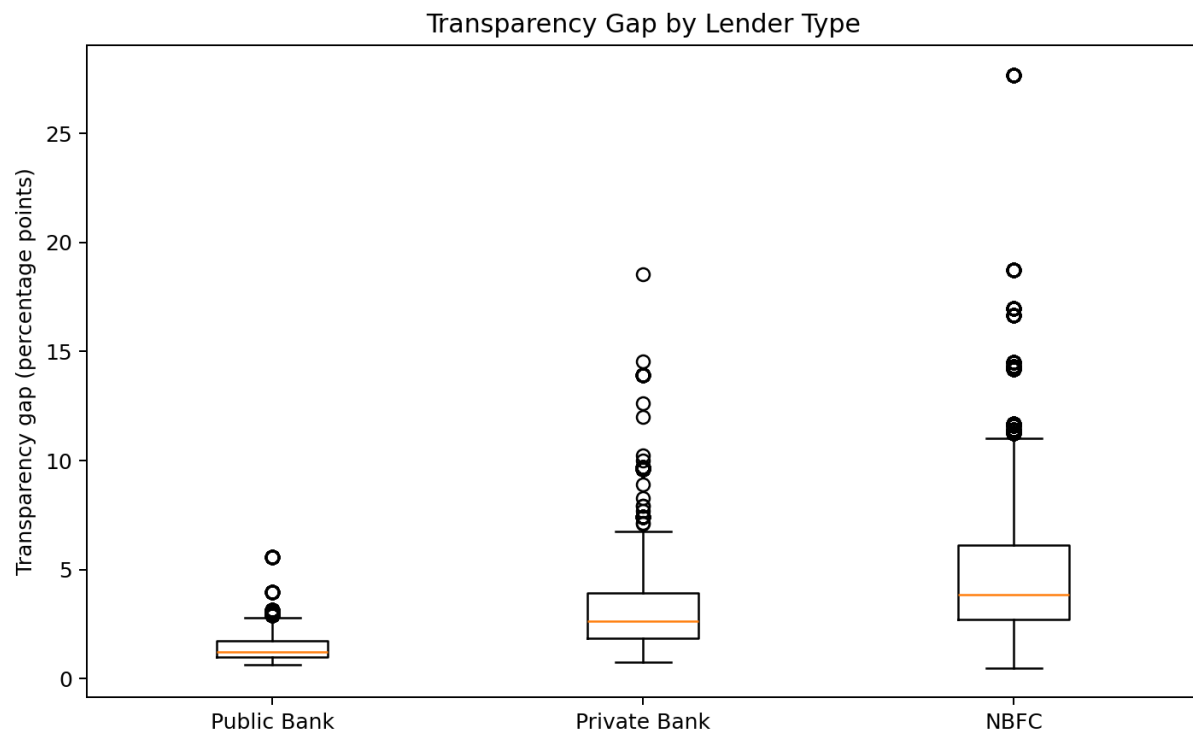
The studies are not directly score-comparable because their datasets and objectives differ. The useful comparison is methodological: this project combines public market terms, financial cash-flow calculations and a reproducible lender-level scoring framework.

## 6. Results, Business Insights and Recommendations

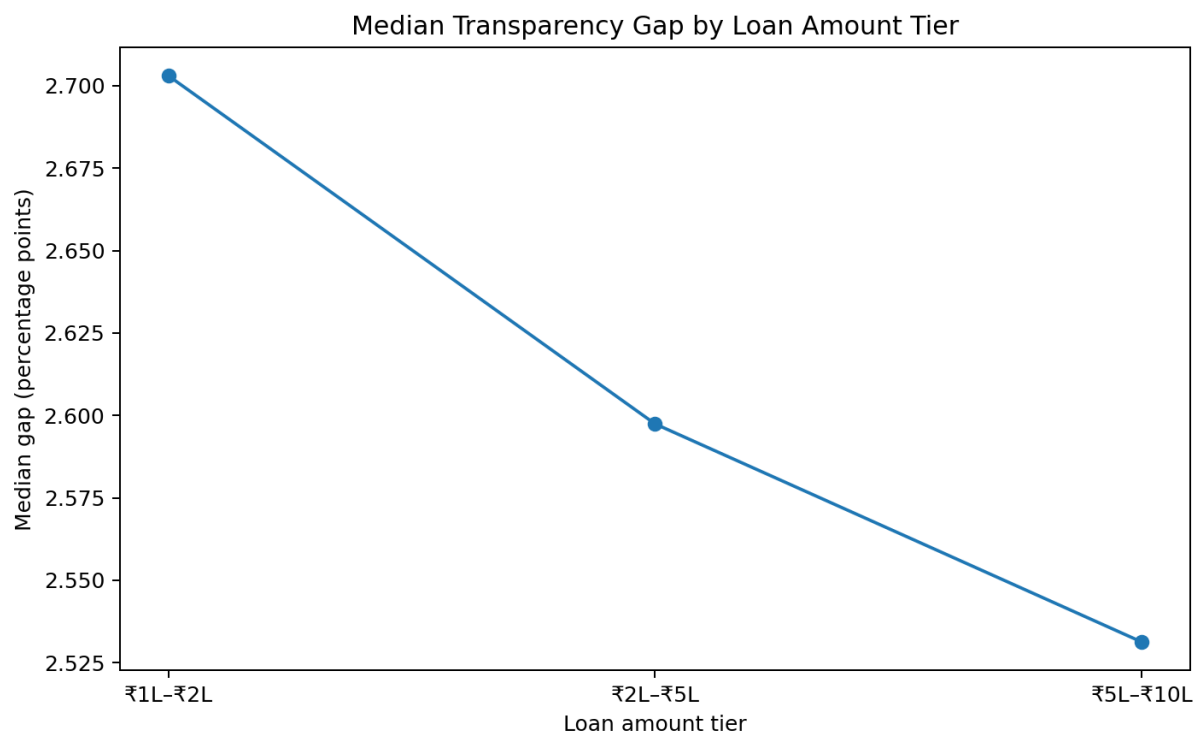
### Lender-type comparison

| Lender type  | Lenders | Mean gap (pp) | Median gap (pp) | Mean APR (%) | Median APR (%) |
|--------------|---------|---------------|-----------------|--------------|----------------|
| NBFC         | 14      | 5.04          | 3.87            | 16.62        | 16.08          |
| Private Bank | 8       | 3.32          | 2.63            | 13.58        | 13.07          |
| Public Bank  | 10      | 1.48          | 1.24            | 11.29        | 11.21          |

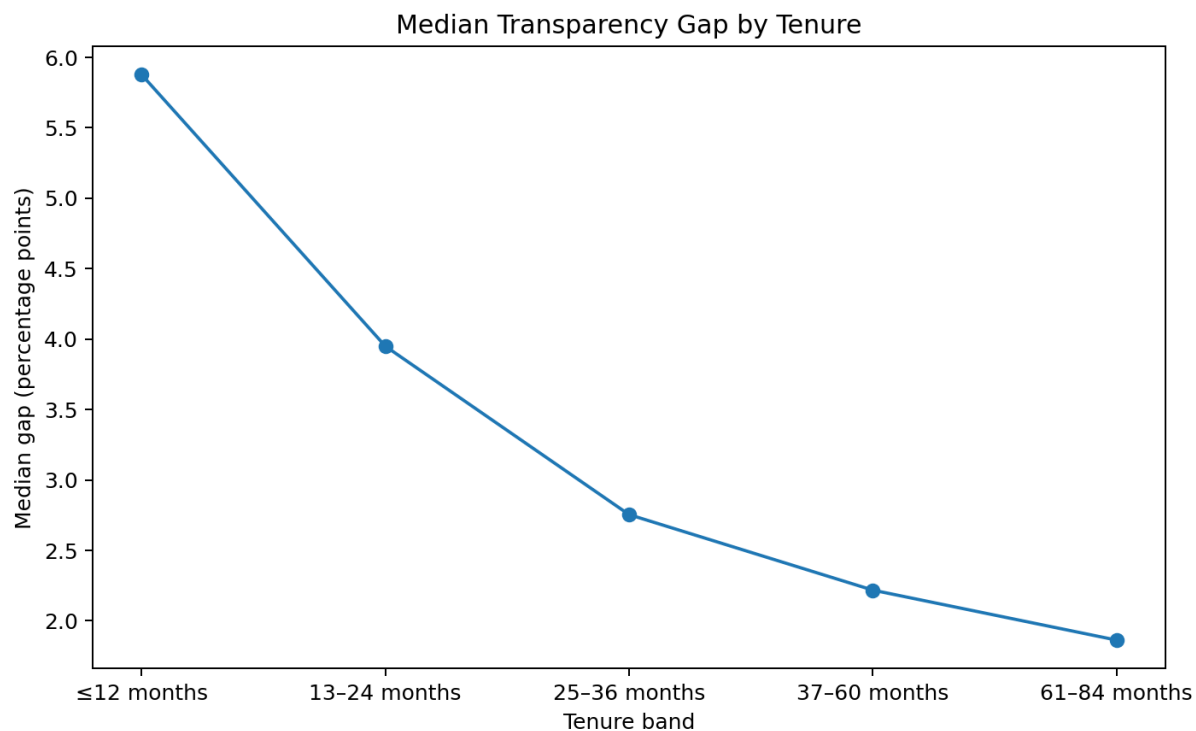
Lender-level Kruskal-Wallis:  $H=18.407$ ,  $p=0.000101$ . This indicates different median-gap distributions across lender-type groups in this dataset; it does not establish causality or generalize to every Indian lender.



## Loan amount and tenure effects



One-time fees can represent a larger proportion of smaller sanctioned amounts. Consequently, the same fee rule can have a different annualized impact across loan sizes.

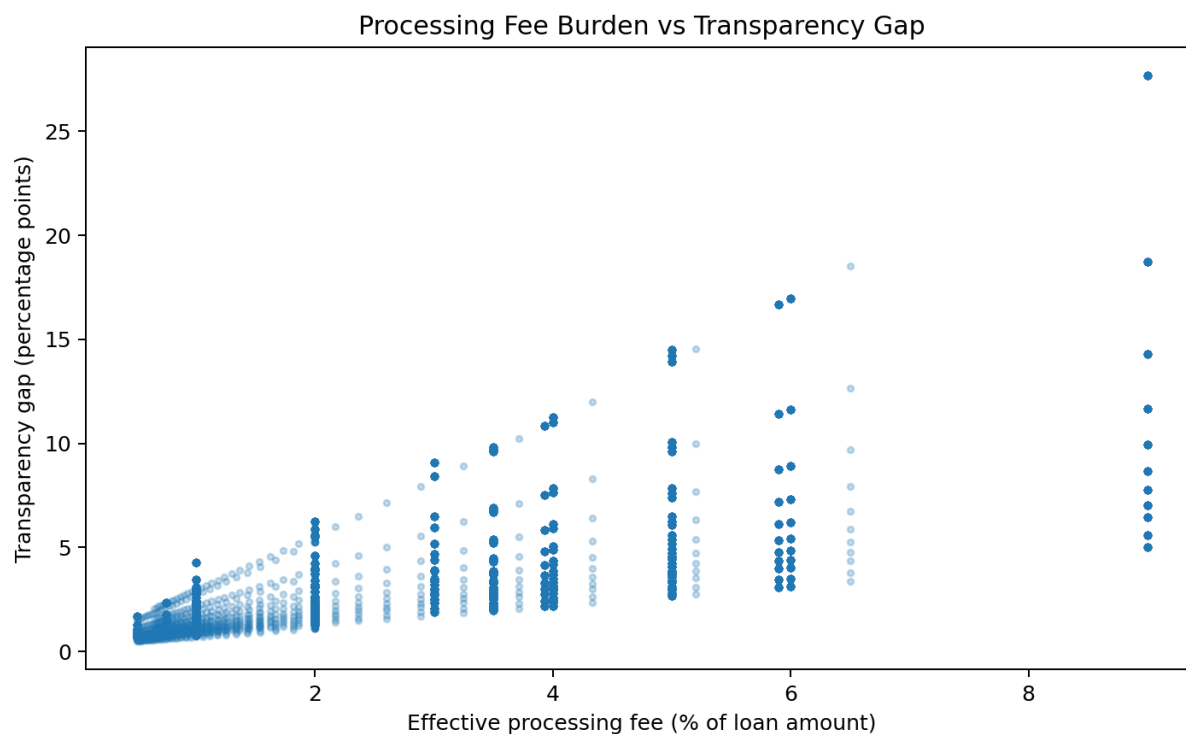


Shorter tenures can amplify the annualized effect of one-time charges because the upfront cost is spread over fewer months.

## Exploratory regression

| Variable                     | Coefficient | p-value    |
|------------------------------|-------------|------------|
| Intercept                    | 1.5587      | 1.733e-05  |
| advertised_rate_pct          | 0.1605      | 2.611e-18  |
| processing_fee_pct_effective | 1.1393      | 4.126e-225 |
| np.log(loan_amount_inr)      | 0.0082      | 0.1348     |
| tenure_months                | -0.0706     | 2.787e-13  |

The model  $R^2$  is 0.728. Processing-fee burden has coefficient 1.139; advertised rate has coefficient 0.161; tenure has coefficient -0.071. These are associations in the standardized scenario data, not causal effects.



## Recommendations

- Comparison platforms should display advertised rate, net amount received, processing fee + GST, EMI, total repayment and effective APR together.
- Borrowers should compare total cash received and total repayment rather than headline rate alone.
- Lenders can improve transparency by clearly separating headline pricing from mandatory upfront charges.
- Audit teams can use standardized scenarios to screen for large gaps, then inspect the underlying Key Facts Statement.
- Analysts should refresh the source dataset because public rates and fees can change.

## 7. Conclusion and References

The study provides a reproducible framework for translating public personal-loan pricing terms into standardized effective borrowing-cost scenarios. Across 13,024 scenarios derived from 32 lender records, the mean gap is 3.50 percentage points and the median gap is 2.58 percentage points. The results demonstrate why headline rate alone is insufficient for comparing the modeled cost of credit when upfront charges are present.

The main limitation is that the dataset is scenario-based rather than borrower-level transaction data. Public rates are indicative and actual offers vary with credit profile and eligibility. The transparency tiers are analytical categories created for this case study and should not be presented as legal or regulatory classifications.

### References

1. Paisabazaar. (2026). Personal Loan: Interest Rates, Loan Amount, Tenure and Processing Fees. <https://www.paisabazaar.com/personal-loan/>
2. Paisabazaar. (2026). Personal Loan Processing Fees. <https://www.paisabazaar.com/personal-loan/personal-loan-processing-fees/>
3. Tantri, P., & Vishen, S. (2025). Does transparency about banks' lending costs lower firms' borrowing costs? Evidence from India. *Journal of Accounting and Economics*, 79(2–3), 101737. <https://doi.org/10.1016/j.jacceco.2024.101737>
4. Ali, A., & Marisetty, V. B. (2026 issue). Are FinTech lending apps harmful? Evidence from user experience in the Indian market. *British Accounting Review*, 58(3), 101269. <https://doi.org/10.1016/j.bar.2023.101269>
5. Sidharta, Y., Nurdina, N., & Putri, N. M. (2024). The Effect of Interest Rate, Administrative Fees, and Risk on Online Lending Decisions on Fintech Lending Applications. *IJEBar*, 8(4). <https://doi.org/10.29040/ijebar.v8i4.16178>

Course submission alignment: README, dataset, notebook and final report are included. Verify the student identity fields and source accessibility before committing to GitHub Classroom.